Q.28 Explain the on-off controller & write 2 disadvantage.

Q.29 Discuss the Diaphragm operated valve.

Q.30 Explain the Analog and digital modulation.

Q.31 Draw the Block diagram of AM transmitters.

Q.32 Explain DSB-SC and draw its spectrum.

Q.33 Discuss the principle of working of the following ASK.

Q.34 Explain the spread Spectrum Techniques.

Q.35 Draw the block diagram of FM receiver.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain the generalized block diagram of a process control system.

Q.37 Explain how a PID controller works by combining P, I and D action. Why is PID control widely used in industrial applications.

Q.38 Explain the principle of Frequency Shift Keying (FSK) modulation. How does it differ from other digital modulation schemes like Amplitude Shift Keying (ASK) and Phase Shift keying (PSK).

(20)

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Roll No. ....

5th Semester/ Mechatronics
Subject : Process Control & Data Communication

Time : 3 Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 A system that does not use feedback is called

a) Closed-loop system b) Open loop system

c) Linear system d) Non linear system

Q.2 A system is said to be time invariant if:

a) Its output changes with time

b) Its behaviour changes when input is shifted in time

c) Its behaviour remains the same when input is shifted in time

d) It produces sinusoidal output for sinusoidal input

Q.3 Which system has better accuracy and automatic correction

a) Open loop system b) Closed loop system

c) Both A & B c) None of the above

Q.4 Which instrument is commonly used to measure flow

a) Thermocouple b) Orifice meter

c) Bourdon tube d) Float switch

Q.5 What type of signal is typically used in industrial process control systems

a) 0-5V DC b) 4-20 mA

c) 10-100V AC d) 0-10V AC

Q.6 What does the 'P' in a PID controller stand for

a) Process b) Pressure

c) Proportional d) Phase

Q.7 What is the main disadvantage of using only a P controller

a) Slow response

b) Overshoot

c) Steady-state error remains

d) Requires derivative gain

Q.8 A solenoid value operates using which principle

a) Hydraulic force

b) Pneumatic pressure

c) Electromagnetic actuation

d) Mechanical linkage

Q.9 In Amplitude Modulation (AM), which parameter of the carrier signal is varied

a) Frequency b) Amplitude

c) Phase d) Bandwidth

Q.10 What does DSB-SC stands for

a) Double Sideband Standard Carrier

b) Double Sideband Suppressed Carrier

c) Dual Sideband Single Carrier

d) Dual Sideband Super Carrier

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

Q.11 ASK stand for_______.

Q.12 Write on disadvantage of an open-loop system is.

Q.13 The error signal is the difference between _________.

Q.14 Write unit of pressure.

Q.15 Which sensor is used to measure temperature in industrial processes.

Q.16 What does the 'P' in PID controller stand for.

Q.17 Name one common detector used in AM receivers

Q.18 How many bits are transmitted per symbol in QPSK.

Q.19 Write name of modulator is used in FM transmission.

Q.20 What is the main purpose of using frequency hopping.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Write five difference between open loop and closed loop control systems

Q.22 What are the main components of a closed-loop control system?

Q.23 Define the process variable (PV)? Give examples.

Q.24 What is Set point (SP)? How does it relate to Process variable?

Q.25 Explain PID controller? Define P, I and D actions.

Q.26 Discuss the Proportional controller? Explain its working principle.

Q.27 Explain the Relative merits and demerits.